Structures and Architecture. REstructure REmaterialize REthink REuse - Rinke & Frier Hvejsel (Eds)
© 2025 The Author(s), ISBN 978-1-041-11139-9
Open Access: www.taylorfrancis.com, CC BY-NC-ND 4.0 license

Exploring daylight on northern latitudes — Assessing quantitative and qualitative aspects in educational practices

K. Næss
Institute of Architecture, The Oslo School of Architecture and Design (AHO), Norway

A.J. Petersen
Department of Building and Environmental Technology, The Norwegian University of Life Sciences

ABSTRACT: Daylight significantly affects behavior, well-being, and perception of room quality. While daylight principles are well understood in continental Europe, the Nordic climate, with its low-angle sun and diffuse light, requires a distinct architectural approach. Despite growing interest, there is a dearth of structured application of methods for assessing daylight quality and quantity in the process of making.To address this need, a master's elective course on daylight at the Oslo School of Architecture has been developed. This paper presents our pedagogical approach to integrating and embedding the scientific aspects of daylight into the creative process to meet the need for a specialized approach to designing buildings at higher latitudes. The course equips students with a deep understanding of Nordic light characteristics, daylight theory, and cultural heritage, emphasizing the roles of light and shadow in architecture. A hands-on approach, using physical and digital models, allows students to explore daylight interactions with space, structure, and materials.

1 INTRODUCTION

The importance of daylight with regards to biological aspects as well as how important daylight is to architecture, is well established (Dubois et al., 2019). Daylight is a scarce resource in the Nordic regions, due to the significant variations in day length and extended periods of dusk and dawn. Additionally, there are substantial differences in local climate variations in topography and landscape. Given that overcast weather predominates at these latitudes, diffuse light can be considered the defining characteristics of the Nordic light (Mathiasen, 2015).

The unique aspects of daylight at high latitudes necessitates an awareness of local conditions when considering daylight in architecture (Mathiasen, 2015). Despite globalized trends in our relationship to the outdoors through apertures, daylight conditions remain intrinsically tied to a specific geographic place and latitude. This recognition of the site-specific nature of daylight, can be traced back to Vitruvius in *Ten Books of Architecture* (Pollio & Gwilt, 2015). In a Nordic context Christian Norberg-Schultz emphasized the significant influence of local environment on traditional architecture, and advocated that daylight stands as one of the most important components in shaping both space and place (Norberg-Schulz, 1979, 1993). Additionally, utilization of daylight in our buildings is heavily dependent on room proportions and geometry as well as placement and form of windows (Mathiasen et al., 2022).

Historically, using tools to assess daylight in buildings has been within the architect's domain, a tradition dating back to Vitruvius. Architects such as Alvar Alto, Sigurd Lewerentz and Sverre Fehn, provide good examples of precision in designing buildings specifically for the Nordic light. For these architects, daylight was as an integral part of the design process,

DOI: 10.1201/9781003658641-182

and a deep understanding of daylight was in their time a natural part of the architects' tools and methods for making architecture.

In recent years, there seems to be a noticeable dearth in architectural education regarding the methods and tools needed to assess both the quantity and quality of daylight during the process of Making. This paper aims to address this gap by presenting a comprehensive pedagogical approach to teaching Nordic light in architecture, which has been tested and evaluated over multiple semesters at the Oslo School of Architecture.

2 THE IMPETUS FOR A COURSE ON DAYLIGHT

Daylight is so fundamental to architecture that it is almost taken for granted, potentially leading to a glaring omission from the curriculum of architecture schools. *In general, expertise on daylight is rare and requires many years to acquire. Whereas this knowledge used to be a part of the education of an architect, daylight expertise has in recent years lost importance* (Dubois, M-C, 2019). Tacit knowledge, traditionally passed down through generations of architects, often lack the precision, tools, and methods needed to accurately describe the role and quality of daylight in architecture. This has led to discrepancy in architectural representation, where what is envisioned in the design stage often fails to capture true daylight conditions in situ.

Engineering research on daylight typically focuses on the quantitative (objective) aspects, whereas architects have traditionally been trained to emphasize the qualitative (subjective) dimensions. However, when both methods are presented, students tend to prefer quantitative, measurable methods. This can indicate that quantitative methods are easier to assess and evaluate, although qualitative should be more familiar in the architects´ domain. To reduce dependence on external expertise and avoid an over-reliance on quantitative methodologies, architects must develop proficiency in both the *quantitative* and *qualitative* aspects of daylight. Moreover, architects need to master essential, analytical tools that enable a conscious use of daylight in the early design processes. Mastering both perspectives is crucial for the relevance of the architect in the future.

This framework has guided our approach, and the primary objective of the course has been:

> to equip our students with tools and methods from both a quantitative and qualitative perspectives, as well as educating them in architecture with higher awareness of the importance of daylight, thereby strengthening their position in further architectural practices.

In general, architectural education at AHO emphasize the importance of working with structure and daylight. The Oslo School of Architecture bridges a polytechnical approach with a handicraft tradition. This course is part of the Technology and Practice section, which aims to maintain a strong connection to the practicing field. However, a scientific approach to working with daylight has not been fully integrated in the creative *Processes of Making* within our architectural education until the introduction of this course. Previously, the focus tended to be either scientific or artistic, emphasizing either knowledge or intuitive explorations. In an effort to merge the scientific aspects of daylight with the more intuitive and diverse approaches to its qualitative aspects, we have collaborated with other researchers in the field of daylight studies in the Nordic Region. The course has been developed in collaboration with researchers at Cluster for Light, Royal Academy Copenhagen and BUILD, Aalborg University. With contributions from researchers at The KTH Royal Institute of Technology, Lunds Tekniske Högskola and The Norwegian University of Life Sciences.

3 LEARNING GOALS

A fundamental discussion within architectural education, is how to balance the measurable and the immeasurable, the objective and the subjective — and *Knowledge and Knowing,* where *Knowing* is freely interpreted from Kahn as the intuitive exploration in a creative process

(Kahn, 2003). Kahn states that knowing is based upon one´s own feeling and experience and knowledge is something that is learned. "*. . .a concentration of knowing must not be used as an excuse to underestimate the value of learning, in that one may also make discoveries through acquired knowledge*" (Fjeld, 2020. p.182). To meet common academic teaching challenges of discrepancy in course description and evaluation, there has been a focus on a constructive alignment in the course to achieve deeper learning and to include all students (Biggs, 1996, 2012).

Within the framework of the studio, the objective is to encompass all perspectives, arguing that *knowledge* should not be a hinder for *knowing, and vice versa. It is recognized* that the creative process of incorporating daylight as a main premise does not necessarily derive from precise tools used to draw, measure, or represent light in the context of architecture. To gain a deep understanding of how daylight works, learning activities are developed to directly integrate the scientific aspects of daylight into the creative process. This approach serves as a method to practice and balance between *knowledge and knowing —and knowing* and *knowledge* (Fjeld & Fjeld, 2019).

The specific learning outcomes that the course aims to fulfil are as follows; the students should:

- be able to translate theory and methods of daylight into to an architectural process of making.
- be able to make architecture with daylight as the main impetus.
- be able to measure and represent daylight with both digital and analogue tools.
- be able to simply, but accurately, sketch daylight by hand.
- have an intimate knowledge of the character of daylight at high latitude and climate.
- have an overview of the history of daylight and apertures in architecture at high latitudes.
- have the necessary knowledge of building regulations and requirements.

By the end of the semester, students should be equipped with the ability to consciously assess both quantity and quality of daylight in their projects, by using both digital and analogue tools to measure and represent daylight. They should develop an inherent understanding of how proportions and geometry of a room, as well as geometry and placement of light-opening(s), affects daylight condition within a space. As an overarching learning outcome, they should also gain a new awareness of the diverse and specific aspects of daylight in Nordic architecture.

In various iterations of this course, we have experimented with initiating the learning process through intuitive exploration (*knowing)*, before introducing structured methods and analytical tools. We found this to be an effective way of anchoring *knowledge* in prior explorative experiences. However, while beneficial, this sequence — from intuitive knowing to structured knowledge— demands more time for the various elements of *knowledge* to settle as *knowing*. Consequently, it is a challenge to facilitate deeper learning within the timeframe of an elective course and may be better suited for a studio course.

We therefore generally structure the course by initially emphasizing structured *knowledge*, providing a theoretical basis for an independent study of daylight. This approach facilitates *knowing*, anchoring the *knowledge* to ensure deeper learning and alignment with the stated learning outcomes.

4 THE COURSE STRUCTURE AND METHODS

The master-level semesters at AHO are organized so that each student participates in a master-level studio, as well as an elective course. Studio work typically involves an in-depth design of a building and/or landscape project, supervised by a team of professors and lecturers. Consequently, students enrolled in the daylight elective are simultaneously engaging deeply with their own design project. As the elective is open to all master-level students, participants come from various studios and work on a diverse range of projects. The daylight

elective usually has around 20 students, with a total of 14 working days (equivalent to 98 hours) dedicated to the course.

Recognizing the pedagogical challenges in balancing *knowledge and knowing* — the course is structures into two Modules that reflect the intended learning outcomes: *Module 1: Tools & Methods (knowledge)* and *Module 2: Work (knowing)*. This structure is the result of critical evaluations involving students feed-back and refinement for four semesters. Essential for the pedagogy is facilitating a transition from learning theory, tools and methods — to a deeper understanding that occurs when students embed these tools in their own projects.

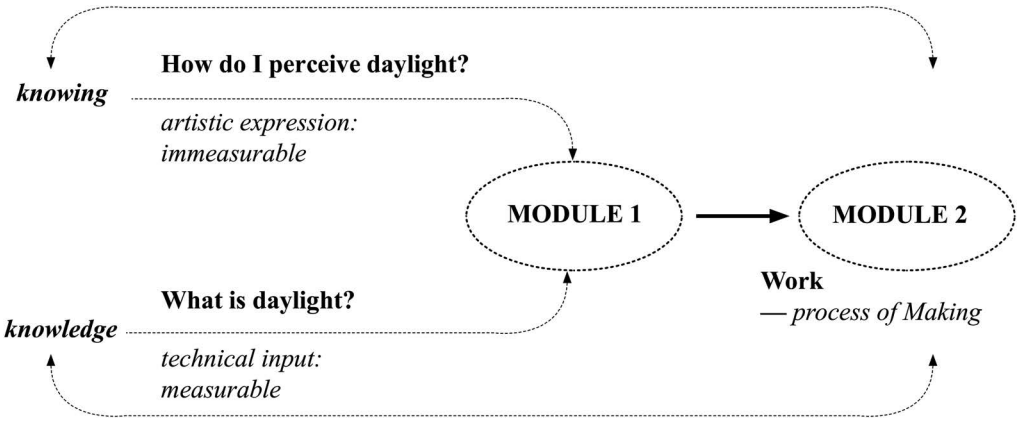


Figure 1. Course structure, organized in two modules, anchoring course alignment and embedding knowledge in knowing.

4.1 *Module 1 – Tools & Methods (Knowledge)*

In Module 1, students engage in a multifaceted exploration of daylight in architectural design. The objective of Module 1 is to provide students with foundational knowledge on daylight theory, along with the tools to draw, measure and represent daylight. The pedagogical approach emphasizes deeper learning through hands-on methods, encouraging students to use both analog and digital tools to assess quantity and quality of daylight in a defined room. Additionally, students gain firsthand experience of how daylight behaves at different scales.

The assignment in Module 1 is inspired by a teaching method described in Grønlund (2022). This method is based on the idea that students can gain insight into the qualitative perspectives of daylight by observing how it changes within their own home over time, using photography as a tool to represent their perception of daylight in the space. This research-based teaching has been further developed for this course.

As initially outlined in Grønlund (2022), this assignment was mainly developed as a qualitative method that emphasized the experiential aspect of daylight's multiple variations in 1:1. For this course, however, the assignment has been expanded to include a quantitative perspective as well, allowing students to better understand the relationship between the measurable and the intangible aspects of daylight. This expansion has been achieved through direct collaboration with researchers from the engineering field, as well as contribution from artists, photographers, and historians.

Students are introduced to "*the no-sky line*" first described by Vitruvius in (Pollio & Gwilt, 2015), which defines the area in a room where the sky is visible, utilizing the method from (Littlefair et al., 2022; Tregenza & Loe, 2013). They are also taught a sketching method developed in conjunction with this course that defines the "*upper-and-lower daylight limit*" of the daylit zone, illustrating how the accuracy of the latter drawing technique can be verified in their representation of daylight (Petersen, 2025). Students start by measuring and sketching a room in their home, from which they create both physical and digital models. They then

capture images of daylight over time, both in situ and in a 1:20 scale model, as well as in the digital model. Thereafter students conduct daylight simulations on the digital model to measure the daylight quantity, under the guidance of researchers. They also learn to sketch daylight in plan and section, embedding digital simulations in their own artistic expression (drawing technique) under the guidance of artists and drawing teachers. This combination of tools aims to enhance their creative process in their further work.

By the final presentation of Module 1, most students have developed an understanding of the relationship between assessing daylight at 1:1 scale, in 1:20 models used in their process, conducting simulations, and applying 2D drawing techniques. This comprehensive approach enables them to choose between precise digital measurements and analogue sketching techniques to evaluate daylight potential in their own work.

Additionally, throughout the semester, students observe how seasonal changes affect daylight in their room —such as falling leaves, snow accumulation, and shifts in the sun´s angle —providing them with firsthand experience of how external factors impact daylight within a space.

4.2 *Module 2 – Work (Knowing)*

The objective of this module is to apply theory, tools, and method to explore the role of daylight in students´ own architectural production, aiming to integrate the knowledge gained in Module 1, into their individual processes. To this end, students are encouraged to engage in their own creative process of *Making*, grounded in the context of their respective studio courses.

The students will begin by exploring a personal interest in daylight, focusing on how structure, light, and materials interact to create architectural space. Unlike Module 1, where students could physically enter a pre-existing 1:1 room to observe daylight behavior, this module requires them to use their imagination and creativity to develop physical representations of daylight conditions through scale models, drawings, images, and other mediums. In this module, students may choose to initiate their daylight investigations from a quantitative perspective, a qualitative perspective, or a combination of both, depending on their project theme. The only requirement is that they discuss daylight within a physical, spatial context, defined by structure and surfaces with defined material surfaces, orientation, and relevant contextual factors such as obstructions, latitude and topography.

5 LEARNING OUTCOMES & EXAMPLES OF STUDENT WORK

Perhaps the most effective way to evaluate the pedagogical approach is by involving students in forming future semesters and by studying their architectural production. There has been a structured evaluation at the end of each semester. The following sections present key insights from these evaluations, along with some representative examples of student work.

5.1 *Evaluation of the course*

At the beginning of the semester, students are introduced to clearly defined learning outcomes, which are assessed at the semester´s end through an evaluation where students articulate their learning experiences. In this process, students collaboratively create a report detailing *what they have learned, what they liked, what they lacked, and what they could do without* (Bjælde et al., 2023). The objective is to foster a reflective process among the students while supporting the iterative planning of future courses.

Som examples of feedback received are as follows:

> *"I learned how to work with light and to understand how light works by using analysis with simple hand drawings. After what I have learned I would not do the same now as I did last semester. This was a fast and simple drawing tool that will be nice to use in future projects.""I learned to look at the models differently and the models got another meaning. Before (this course) it was more like an object, and I was fooled to think it is a good light condition, but the model had for instance open walls and could not be used to assess daylight. It has changed the perspective of the model as an instrument to investigate light."*

"It has been an eyeopener to see how light enters the room, it is interesting to now see the no-sky line (upper and lower daylight limit) everywhere. It is simple, but something we never have been introduced to before that is very important in architecture."

5.2 *Examples of student work*

By adding 2D drawing techniques to their own observations of daylight captured through photography, students observe how the "*upper and lower daylight limit*" is "*surprisingly accurate*" with their observations in 1:1. An example of student work on this theme is featured in *Figure 2,* where a student compares the simple drawing techniques from a section drawing with in situ photos of their own room.



Figure 2. An example of student work, where the upper and lower daylight limit as well as the no sky line are drawn in a section on the left and compared to real life conditions on the right. The apparent upper daylight limit is drawn in a dashed line in the in-situ photo. (Photo: Martine Beitohaugen).

Student work from the workshop "*Drawing Light — from a technical and artistic perspective*" inspired by the concept of Light zone(s), described in Madsen (2015), demonstrates how digital daylight simulations can be interpreted through individual experience and spatial perception. Various hatching techniques were employed to blend technical knowledge with artistic expression, ensuring its inclusion in future studies (Module 2). An example of student work from this workshop, illustrating a representation of daylight conditions based on both daylight simulations and drawing techniques, is shown in Figure 3.

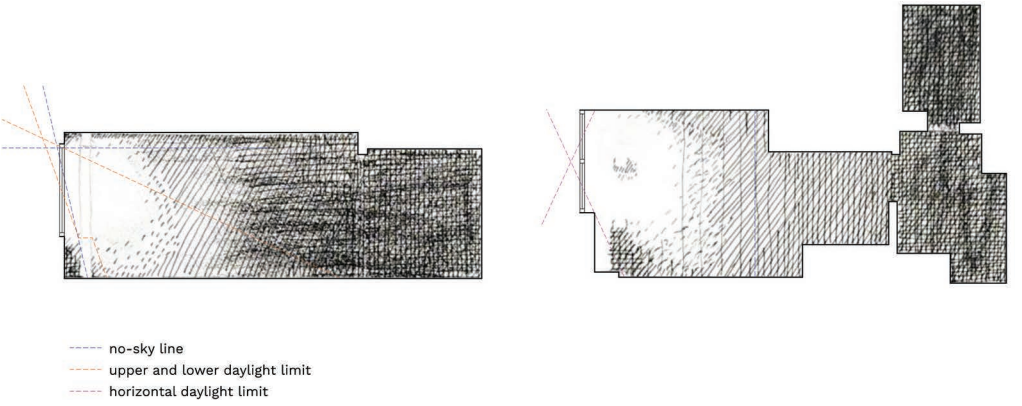


Figure 3. An example of student work from the drawing workshop, showing the "light zone" in a section as well as a plan. This representation is based on a combination of daylight simulations as well as analogue drawing techniques. (Drawing: Malte Grannemann).

By constructing a scale model and positioning it within the actual room it represents, students directly observe how daylight behavior at different scales. They assess daylight conditions at a scale 1:1 and at a 1:20 scale, utilizing a daylight simulator that approximates overcast light conditions. This comparison effectively integrates tools that combine individual experience with analytic techniques, fostering deeper learning. An example illustrating student work that compares actual in-situ conditions under an overcast sky to a scale model photographed in a light simulator can be seen in Figure 4.

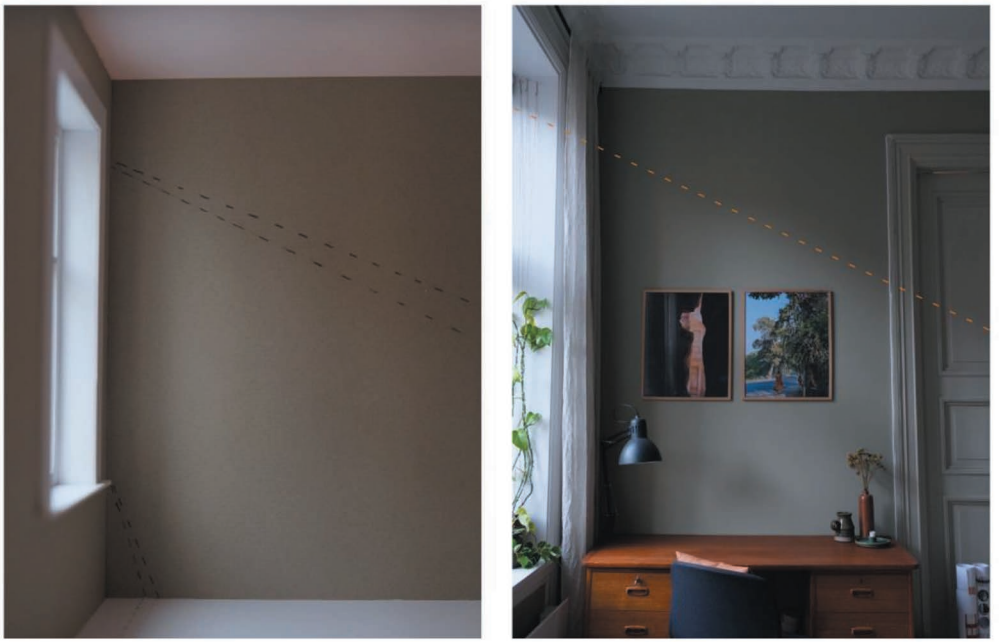


Figure 4. Light conditions in situ in a student apartment compared to a 1:20 model in a light simulator. The apparent upper daylight limit is drawn in a dashed line on both photos. In situ on the on the right and model on the left. [Photo: Martine Beitohaugen].

6 CONCLUSION

The main pedagogical challenge lies in embedding the *knowledge* of what daylight is into the *knowing* of how to work with daylight in architecture. While recognizing the difficulty in fully integrating tools and methods into a creative process within the limited time frame of 14 working days over a semester. Nevertheless, final submissions at the end of the semester indicate that students gain a deeper understanding daylight´s potential within their projects. Furthermore, we observe that many students continue to apply these tools later in their subsequent studies.

Despite of general challenges of integrating technical aspects in the artistic field, and vice versa, we observe a growing interest among students in working with both dimensions within architecture. This interest may stem from changes in the climate, that again makes students more aware of how climatic changes affects buildings and with that the discourse in the architectural field.

After four semesters of iteration, we observe that both the course structure and the embedded research evolve through the iterative planning, continuously enhancing the teaching in daylight as part of the main curriculum at the Institute of Architecture. The

collaborative efforts of the researchers involved in the course support small adjustments each semester, allowing for the integration of new research and new collaborative efforts that strengthens students´ precision in working with daylight. This interdisciplinary collaboration enriches the assignment, providing students with a more comprehensive understanding of daylight in architectural design. We believe this approach holds significant potential for advancing architectural education and with that improving the overall quality of daylight in our built environment which in turn is good for health and well-being.

ACKNOWLEDGEMENTS:

The authors would like to extend special thanks to Louise Grønlund and Nanet Mathiasen for sharing their research and generously assisting in the development of this course at AHO. We also wish to thank Malin Alenius, Leif Houck, Sareh Saeidi, Tobias Kristiansen, Marie-Claude Dubois, Katja Bülow, Cathrine Vigander and Tine Hegli for their invaluable support in establishing and advancing research-based teaching in daylight at the Institute of Architecture, the Oslo School of Architecture and Design (AHO).

REFERENCES

Bjælde, O.E., Boud, D., & Lindberg, A. B. (2023). Designing feedback activities to help low-performing students. *Active Learning in Higher Education*. https://dx.doi.org/10.1177/14697874231212820

Dubois, M.-C., Gentile, N., Laike, T., Bournas, I., & Alenius, M. (2019). *Daylighting and lighting under a Nordic sky* (Vol. 1). Studentlitteratur AB.

Fjeld, P. O. (2020). *The Power Of Circumstance - Architecture And Creative Independence, Six Lectures By Per Olaf Fjeld*. Architectural Publisher B.

Fjeld, P. O., & Fjeld, E. R. (2019). *Louis I. Kahn - The Nordic Latitudes*. University of Arkansas Press.

Grønlund, L. (2022). The Appearances of Daylight – an educational method for studying daylight. *IOP Conference Series: Earth and Environmental Science*, *1099*(1). https://dx.doi.org/10.1088/1755-1315/1099/1/012021

Kahn, L. I. (2003). *Louis Kahn: Essential Texts* (R. Twombly, Ed.). W. W. Norton & Company.

Littlefair, P. J., King, S., Howlett, G., Ticleanu, C., & Longfield, A. (2022). *Site layout planning for daylight and sunlight A guide to good practice from S&P Global*.

Madsen, M. (2015). Light-zone(s): as Concept and Tool. *ARCC Journal*, *4*(1), 50.

Mathiasen, N. (2015). *Nordisk lys - og dets relation til dagslysåbninger i nordisk arkitektur*. Det Kongelige Danske Kunstakademis Skoler for Arkitektur, Design og Konservering.

Mathiasen, N., Grønlund, L., Frandsen, A. K., & Harild, M. (2022). Window geometry and its effect on the experience of illuminated spaces - A study of three daylit architectural cases. *IOP Conference Series: Earth and Environmental Science*, 1099(1). https://dx.doi.org/10.1088/1755-1315/1099/1/012019

Norberg-Schulz, C. (1979). *Genius Loci: Towards a Phenomenology of Architecture*. Rizzoli.

Norberg-Schulz, C. (1993). *Nattlandene : om byggekunst i Norden*. Gyldendal.

Petersen, A. J. (2025). Shedding light on architecture - sketching daylight under an overcast sky. In *Proceedings of the sixth International Conference on Structures and Architecture (ICSA 2025)*. CRC Press / Balkema (Taylor & Francis Group).

Pollio, M. V., & Gwilt, J. (2015). The Architecture of Marcus Vitruvius Pollio: In Ten Books. In *The Architecture of Marcus Vitruvius Pollio: Vol. VI*. Cambridge University Press. https://dx.doi.org/10.1017/CBO9781107280670

Tregenza, P., & Loe, D. (2013). *The design of lighting*. Routledge.